TEST REPORT

For EMC

Report No. ....:	**CHTEW21100066** Report verification :
Project No. ……………….…………..:	**SHT2107068302EW**
Applicant's Name.......................... :	**Chengdu Ebyte Electronic Technology Co., Ltd.**
Address... :	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Test Item Description .... :	**SPI SMD LoRa Module**
Trade Mark...................................... :	EBYTE
Model/Type Reference .... :	E22-900M22S
Listed Model(s) .... :	-
Standard .... :	**ETSI EN 301 489-1 V2.2.3 (2019-11)** **ETSI EN 301 489-3 V2.1.1 (2019-03)**
Date of Receipt of Test Sample……:	Jul.26, 2021
Date of Testing………………………..:	Jul.26, 2021- Oct.14, 2021
Date of Issue…………………………..:	Oct.15, 2021
Result ....……………………..:	**PASS**

Compiled by (position+printedname+signature).. :	File administrators Echo Wei	Echo Wei
Supervised by (position+printedname+signature)... :	Project Engineer Kiki Kong	Kiki Kong
Approved by (position+printedname+signature)... :	RF Manager Hans Hu	Hans Hu

Testing Laboratory Name .... :	**Shenzhen Huatongwei International Inspection Co., Ltd.**
Address... :	1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road, Tianliao, Gongming, Shenzhen, China

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The test report merely correspond to the test sample.

Contents

1. Test Standards and Report Version

1.1. Test Standards

The tests were performed according to following standards:

ETSI EN 301 489-1 V2.2.3 (2019-11)–ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 1: Common technical requirements; Harmonised Standard for ElectroMagnetic Compatibility
ETSI EN 301 489-3 V2.1.1 (2019-03)–ElectroMagnetic Compatibility (EMC)standard for radio equipment and services; Part 3: Specific conditions for Short-Range Devices (SRD)operating on frequencies between 9 kHz and 246 GHz; Harmonised Standard covering the essential requirements of article 3.1(b) of Directive 2014/53/EU

1.2. Report Version Information

Revision No.	Date of Issue	Description
N/A	2021-10-15	Original

2. Test Description

Emission			
Test Item	**Standard requirement**	**Result**	**Test Engineer**
Radiated Emission	EN301 489-1 Clause 8.2	Pass	Haoxin Luo
Conducted Emission(AC Mains)	EN301 489-1 Clause 8.4	Pass	Pan Xie
Harmonic Current Emissions	EN301 489-1 Clause 8.5	N/A	N/A
Voltage Fluctuations and Flicker	EN301 489-1 Clause 8.6	Pass	Pan Xie
Immunity			
Test Item	**Standard requirement**	**Result**	**Test Engineer**
Radio Frequency Electromagnetic Field	EN301 489-1 Clause 9.2	Pass	Jian Li
Electrostatic Discharge	EN301 489-1 Clause 9.3	Pass	Zhenguo Gao
Fast Transients (common mode)	EN301 489-1 Clause 9.4	Pass	Zhenguo Gao
Radio Frequency (common mode)	EN301 489-1 Clause 9.5	Pass	Zhenguo Gao
Voltage Dips and Interruptions	EN301 489-1 Clause 9.7	Pass	Quanhai Deng
Surges	EN301 489-1 Clause 9.8	Pass	Zhenguo Gao

Note: The measurement uncertainty is not included in the test result.

3. Summary

3.1. Client Information

Applicant:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Manufacturer:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Factory:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China

3.2. Product Description

Product Name:	SPI SMD LoRa Module		
Trade Mark:	EBYTE		
Model/Type Reference:	E22-900M22S		
Listed Model(s)::	-		
Power Supply:	DC 5V		
Hardware Version:	V1.0		
Software Version:	V1.0		
Transmitter unit:			
Operation Frequency:	863MHz~870MHz		
Modulation type:	LORA		
Modulation connector:	☒ Without external	☐ External	
Occupied bandwidth[#1]:	125KHz		
Product type:	☒ Wideband Tranceiver	☐ Narrowband Tranceiver	
Antenna type:	Sucker antenna		
Antenna gain[#2]:	3dBi		
Receiver unit:			
Receiver category[#3]:	☒ 2	☐ 1.5	☐ 1

3.3. Accessory Equipment Information

Laptop	
Manufacturer :	DELL
Model No. :	Inspiron 13-5378

3.4. EUT Operation Mode

Test mode O1	Keep the EUT in transmitting with max power.
Test mode O2	Keep the EUT in receiving

Note:
1) * is represent the following meaning in the test report
   O*: Test mode O1, Test mode O2,

Pre-scan above all test mode, found below test mode which it was worse case mode, so only show the test data for worse case mode on the test report.

Test item	Test mode (Worse case mode)
Radiated Emission	Test mode O1
Conducted Emission(AC Mains)	Test mode O1
Harmonic Current Emissions	N/A
Voltage Fluctuations and Flicker	Test mode O1
Radio Frequency Electromagnetic Field	O*
Electrostatic Discharge	O*
Fast Transients (common mode)	O*
Radio Frequency (common mode)	O*
Voltage Dips and Interruptions	O*
Surges	O*

3.5. Modifications

No modifications were implemented to meet testing criteria.

4. Test Environment

4.1. Testing Laboratory Information

Laboratory Name	Shenzhen Huatongwei International Inspection Co., Ltd.
Laboratory Location	1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road, Tianliao, Gongming, Shenzhen, China
Connect information:	Tel: 86-755-26715499 E-mail: cs@szhtw.com.cn http://www.szhtw.com.cn

4.2. Environmental Conditions

During the measurement the environmental conditions were within the listed ranges:

Normal Temperature:	25°C
Lative Humidity	55 %
Air Pressure	989 hPa

4.3. Statement of the Measurement Uncertainty

The data and results referenced in this document are true and accurate. The reader is cautioned that there may be errors within the calibration limits of the equipment and facilities. The measurement uncertainty was calculated for all measurements listed in this test report acc. to CISPR 16 - 4 „Specification for radio disturbance and immunity measuring apparatus and methods – Part 4: Uncertainty in EMC Measurements and is documented in the Shenzhen Huatongwei International Inspection Co., Ltd quality system acc. to DIN EN ISO/IEC 17025. Furthermore, component and process variability of devices similar to that tested may result in additional deviation. The manufacturer has the sole responsibility of continued compliance of the device.

Hereafter the best measurement capability for Shenzhen Huatongweilaboratory is reported:

Test Item	Range	MeasurementUncertainty	Notes
Radiated Emission	30~1000MHz	4.36dB	(1)
Radiated Emission	1~18GHz	5.10dB	(1)
Conducted Disturbance	0.15~30MHz	3.00dB	(1)

(1)This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=1.96

4.4. Equipments Used During the Test

● Conducted Emission							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Shielded Room	Albatross projects	HTWE0114	N/A	N/A	2018/09/28	2023/09/27
●	EMI Test Receiver	R&S	HTWE0111	ESCI	101247	2021/9/14	2022/9/13
●	Artificial Mains	SCHWARZBECK	HTWE0113	NNLK 8121	573	2021/9/17	2022/9/16
●	Pulse Limiter	R&S	HTWE0033	ESH3-Z2	100499	2021/9/13	2022/9/12
●	RF Connection Cable	HUBER+SUHNER	HTWE0113-02	ENVIROFLEX_142	EF-NM-BNCM-2M	2021/9/17	2022/9/16
●	Test Software	R&S	N/A	ES-K1	N/A	N/A	N/A

● Radiated Emission-6th test site							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Semi-Anechoic Chamber	Albatross projects	HTWE0127	SAC-3m-02	C11121	2018/09/30	2022/09/29
●	EMI Test Receiver	R&S	HTWE0099	ESCI	100900	2021/9/14	2022/9/13
●	Ultra-Broadband Antenna	SCHWARZBECK	HTWE0119	VULB9163	546	2020/04/28	2023/04/27
●	Pre-Amplifer	SCHWARZBECK	HTWE0295	BBV 9742	N/A	2020/11/13	2021/11/12
●	RF Connection Cable	HUBER+SUHNER	HTWE0062-01	N/A	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0062-02	SUCOFLEX104	501184/4	2021/02/26	2022/02/25
●	Test Software	R&S	N/A	ES-K1	N/A	N/A	N/A

● Radiated emission-7th test site							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Semi-Anechoic Chamber	Albatross projects	HTWE0122	SAC-3m-01	N/A	2018/09/27	2022/09/26
●	Spectrum Analyzer	R&S	HTWE0098	FSP40	100597	2021/9/13	2022/9/12
●	Horn Antenna	SCHWARZBECK	HTWE0126	9120D	1011	2020/04/01	2023/03/31
●	Broadband Pre-amplifier	SCHWARZBECK	HTWE0201	BBV 9718	9718-248	2021/03/05	2022/03/04
●	RF Connection Cable	HUBER+SUHNER	HTWE0126-01	RE-7-FH	N/A	2021/03/05	2022/03/04
●	Test Software	Audix	N/A	E3	N/A	N/A	N/A

● Harmonic Current Emissions, Voltage Fluctuations and Flicker							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Purified Power Source	California instruments	HTWE0019	HFS500	54513	2021/9/14	2022/9/13
●	Harmonic And Flicker Analyzer	EM TEST	HTWE0018	DPA503S1	0500-10	2021/9/14	2022/9/13
●	Test Software	EM TEST	N/A	DPA	N/A	N/A	N/A

● Electrostatic Discharge							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	ESD Simulator	EM TEST	HTWE0500	ESD NX30.1	11971	2021/07/21	2022/07/20

● Radio Frequency Electromagnetic Field							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Semi-Anechoic Chamber	Albatross projects	HTWE0127	SAC-3m-02	C11121	2018/09/30	2022/09/29
●	Signal Generator	R&S	HTWE0276	SMB100A	114360	2021/08/05	2022/08/04
●	Amplifier	R&S	HTWE0277	BBA150-BC500	102664	2021/08/18	2022/08/17
●	Amplifier	R&S	HTWE0395	BBA150 D400	104197	2021/07/29	2022/07/29
●	Amplifier	R&S	HTWE0396	BBA150 E400	104198	2021/07/29	2022/07/29
●	Power Head	R&S	HTWE0278	NRP18A	101010	2021/08/05	2022/08/04
●	Power Head	R&S	HTWE0389	NRP18A	101386	2021/05/27	2022/05/26
●	Transmit Antenna	Schwarzbeck	HTWE0280	STLP9129	00044	2021/03/30	2022/03/29
●	Field Probe	ETS-LINDGREN	HTWE0321	HI-6153	00130812	2019/05/23	2022/05/22
●	Test Software	R&S	N/A	EMC32	100916	N/A	N/A
○	Audio analyzer	R&S	HTWE3008	UPV	101371	2021/10/09	2022/10/08
○	Radio communication tester	R&S	HTWE0287	CMW500	137688-Lv	2021/9/13	2022/9/12
○	RF Communication Test Set	HP	HTWE0038	8920A	3813A10206	2021/9/13	2022/9/12
○	Digital intercom communication tester	Aeroflex	HTWE0255	3920B	1001682041	2021/9/13	2022/9/12

● Electrical fast transient/burst immunity test, Surge immunity test							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Ultra Compact Simulator	EM TEST	HTWE0004	UCS500M6	0500-19	2021/9/17	2022/9/16
●	Surge Generator	EM TEST	HTWE0003	TSS500M4	1100-04	2021/9/17	2022/9/16
●	3-Phase Coupling Network	EM TEST	HTWE0005	CNI503 S5/16A	0606-01	2021/9/17	2022/9/16
●	Coupling Clamp	EM TEST	HTWE0007	HFK	1501-14	2021/9/17	2022/9/16
●	Test Software	EM TEST	N/A	ISM IEC	N/A	N/A	N/A

● Radio frequency (common mode)							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Signal Generator	IFR	HTWE0022	2023A	202304/060	2021/9/17	2022/9/16
●	Amplifier	AR	HTWE0023	75A250	302205	2021/9/17	2022/9/16
●	6db Attenuator	EMTEST	HTWE0025	ATT6/75	0010230A	2020/10/15	2022/10/14
○	EM Clamp	LÜTHI	HTWE0028	EM101	335625	2021/9/17	2022/9/16
●	Test Software	AR	N/A	SW1004	N/A	N/A	N/A
○	CDN	EMTEST	HTWE0155	CDN M3	0802-03	2021/9/14	2022/9/13

○	CDN	EMTEST	HTWE0154	CDN M2	510010010012	2021/9/14	2022/9/13
○	CDN	EMTEST	HTWE0153	CDN M1/32A	0202-05	2021/9/14	2022/9/13
○	CDN	EMTEST	HTWE0156	CDN M4-N/32A	510010660001	2021/9/14	2022/9/13
○	Audio analyzer	R&S	HTWE3008	UPV	101371	2021/10/09	2022/10/08
○	Radio communication tester	R&S	HTWE0287	CMW500	137688-Lv	2021/9/13	2022/9/12
○	RF Communication Test Set	HP	HTWE0038	8920A	3813A10206	2021/9/13	2022/9/12
○	Digital intercom communication tester	Aeroflex	HTWE0255	3920B	1001682041	2021/9/13	2022/9/12

●	Voltage Dips and Interruptions						
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Purified Power Source	California instruments	HTWE0019	HFS500	54513	2021/9/14	2022/9/13
●	Test Software	California instruments	N/A	CIGUII-5001iX	N/A	N/A	N/A

5. Test Conditions and Results

5.1. EMISSION

5.1.1. Radiated Emission

LIMIT

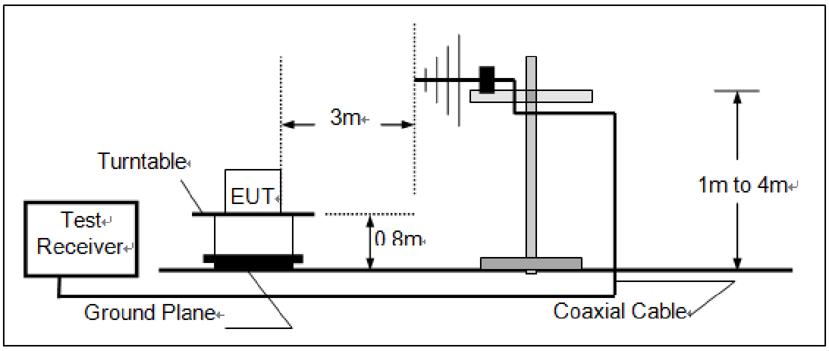
Please refer to ETSI EN301489-1 Clause 8.2.3, Table 4 and CENELEC EN 55032 Annex A Table A.4 & A.5

Frequency range (MHz)	Quasi-peak limitsdBμV/m@3m
30~230	40
230~1000	47

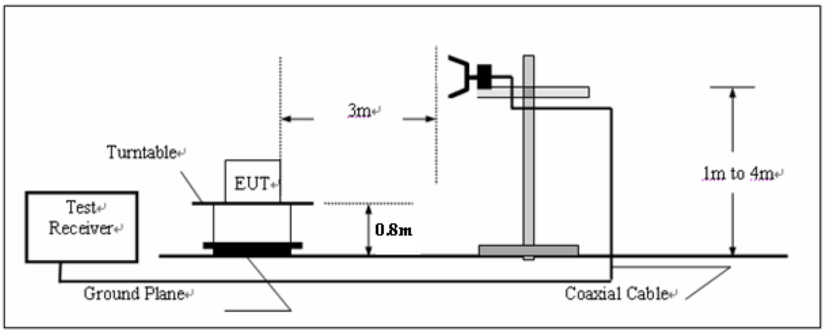
Frequency range (GHz)	Average limitsdBμV/m@3m	Peak limitsdBμV/m@3m
1 ~ 3	50	70
3 ~ 6	54	74

TEST CONFIGURATION

➢ below 1000MHz:



➢ Above 1000MHz



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 8.2.3 and CENELEC EN 55032 Clause 6.3 for the measurement methods

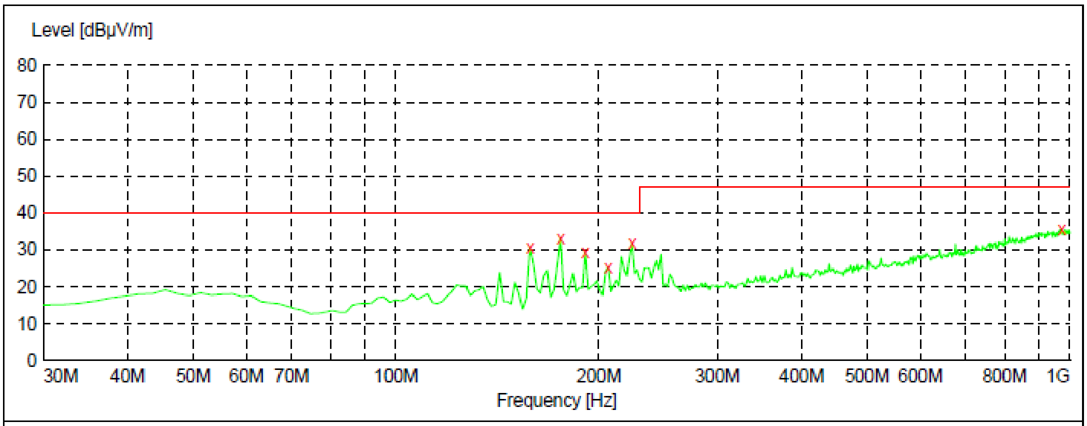
TEST MODE:

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

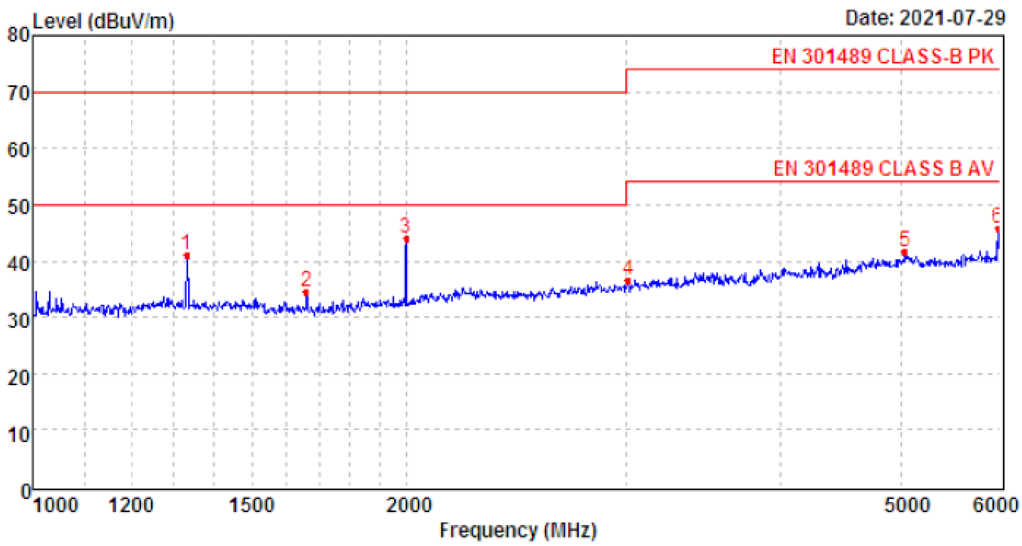
Polarization	Horizontal



MEASUREMENT RESULT: "GM2108316093_red"

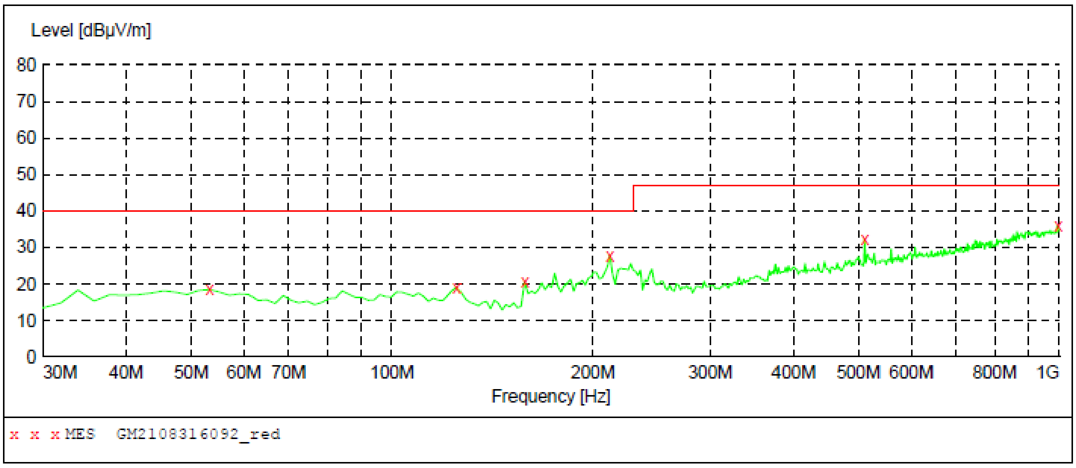
8/31/2021 9:16PM

Frequency MHz	Level dBµV/m	Transd dB	Limit dBµV/m	Margin dB	Det.	Height cm	Azimuth deg	Polarization
158.040000	30.70	-13.5	40.0	9.3	QP	100.0	160.00	HORIZONTAL
175.500000	33.30	-12.6	40.0	6.7	QP	100.0	160.00	HORIZONTAL
191.020000	29.20	-10.9	40.0	10.8	QP	100.0	171.00	HORIZONTAL
206.540000	25.40	-10.5	40.0	14.6	QP	100.0	149.00	HORIZONTAL
224.000000	31.70	-9.7	40.0	8.3	QP	100.0	76.00	HORIZONTAL
970.900000	35.60	8.0	47.0	11.4	QP	300.0	155.00	HORIZONTAL



Mark	Frequency MHz	Reading dBm	Antenna dB	Cable dB	Preamp dB	Level dBm	Limit dBm	Over limit	Remark
1	1329.62	47.27	26.18	4.07	36.37	41.15	70.00	-28.85	Peak
2	1660.42	41.96	25.10	4.56	37.20	34.42	70.00	-35.58	Peak
3	1996.95	50.05	26.09	5.01	37.07	44.08	70.00	-25.92	Peak
4	3015.37	39.21	28.73	6.21	37.46	36.69	74.00	-37.31	Peak
5	5033.76	36.05	32.10	8.85	35.33	41.67	74.00	-32.33	Peak
6	5978.54	38.75	32.46	9.58	35.04	45.75	74.00	-28.25	Peak

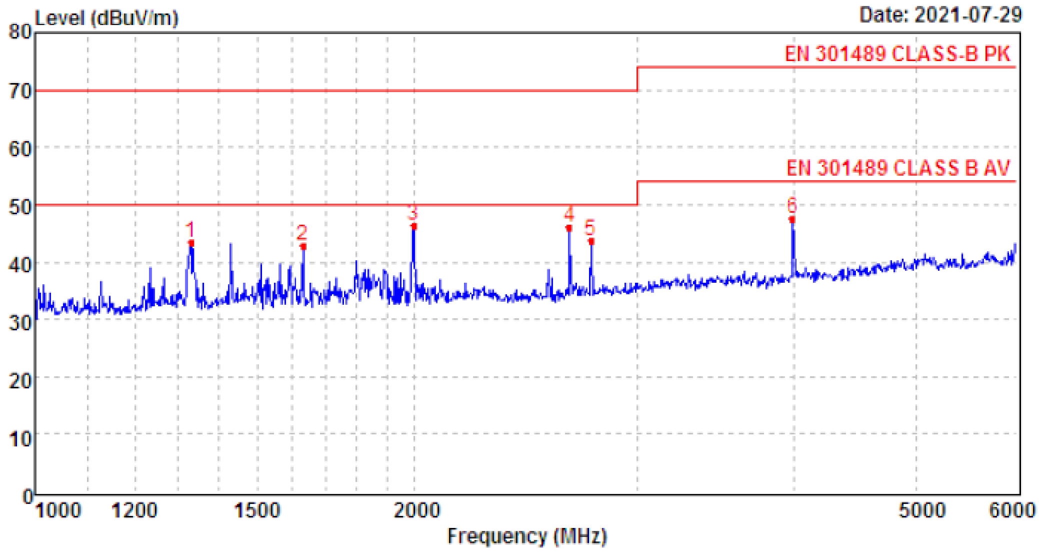
Polarization	*Vertical*



MEASUREMENT RESULT: "GM2108316092_red"

8/31/2021 9:13PM

Frequency MHz	Level dBµV/m	Transd dB	Limit dBµV/m	Margin dB	Det.	Height cm	Azimuth deg	Polarization
53.280000	18.50	-8.9	40.0	21.5	QP	100.0	159.00	VERTICAL
125.060000	19.20	-13.3	40.0	20.8	QP	100.0	122.00	VERTICAL
158.040000	20.60	-13.5	40.0	19.4	QP	100.0	88.00	VERTICAL
212.360000	27.80	-10.5	40.0	12.2	QP	100.0	28.00	VERTICAL
511.120000	32.20	-1.5	47.0	14.8	QP	100.0	206.00	VERTICAL
996.120000	35.90	8.5	47.0	11.1	QP	100.0	171.00	VERTICAL



Mark	Frequency MHz	Reading dBm	Antenna dB	Cable dB	Preamp dB	Level dBm	Limit dBm	Over limit	Remark
1	1329.62	49.51	26.18	4.07	36.37	43.39	70.00	-26.61	Peak
2	1630.93	50.29	25.21	4.51	37.19	42.82	70.00	-27.18	Peak
3	1996.95	52.24	26.09	5.01	37.07	46.27	70.00	-23.73	Peak
4	2655.17	49.64	27.73	5.87	37.01	46.23	70.00	-23.77	Peak
5	2756.98	46.85	28.23	5.95	37.24	43.79	70.00	-26.21	Peak
6	3987.79	46.79	29.90	7.38	36.39	47.68	74.00	-26.32	Peak

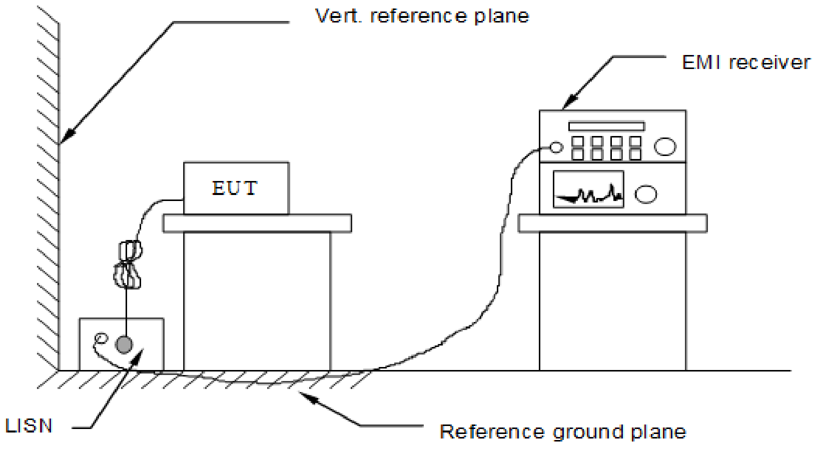
5.1.2. Conducted Emission (AC Mains)

LIMIT

Please refer to ETSI EN301489-1 Clause 8.4.3.2 and CENELEC EN 55032 Annex A3 Table A.10

Frequency range MHz	**Limits** dB(μV)	
	Quasi-peak	**Average**
0,15 to 0,50	66 to 56	56 to 46
0,50 to 5	56	46
5 to 30	60	50
NOTE 1 The lower limit shall apply at the transition frequencies. NOTE 2 The limit decreases linearly with the logarithm of the frequency in the range 0,15 MHz to 0,50 MHz.		

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 8.4.3 and CENELEC EN 55032 Annex A3 Table A.8

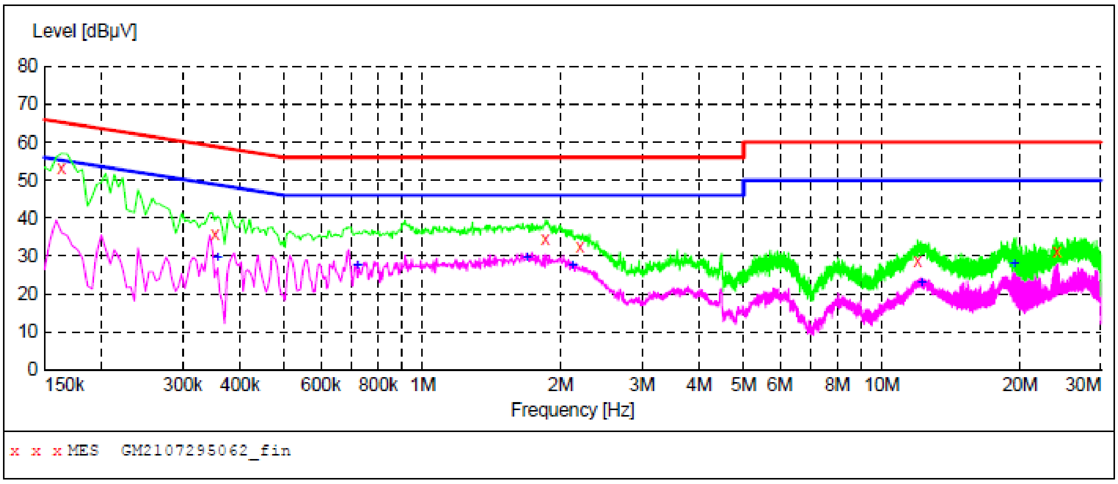
TEST MODE:

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Polarization	L



MEASUREMENT RESULT: "GM2107295062_fin"

7/29/2021 6:27PM

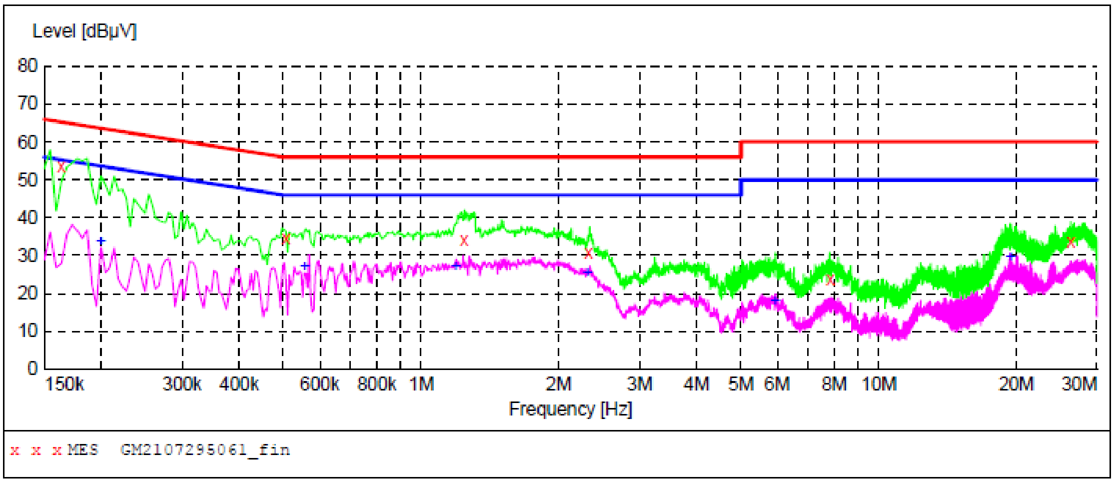
Frequency MHz	Level dBµV	Transd dB	Limit dBµV	Margin dB	Detector	Line	PE
0.163500	53.20	10.2	65	12.1	QP	L1	GND
0.352500	35.90	10.2	59	23.0	QP	L1	GND
1.846500	34.50	10.2	56	21.5	QP	L1	GND
2.197500	32.60	10.2	56	23.4	QP	L1	GND
11.949000	28.70	10.5	60	31.3	QP	L1	GND
24.000000	31.40	10.5	60	28.6	QP	L1	GND

MEASUREMENT RESULT: "GM2107295062_fin2"

7/29/2021 6:27PM

Frequency MHz	Level dBµV	Transd dB	Limit dBµV	Margin dB	Detector	Line	PE
0.357000	29.40	10.2	49	19.4	AV	L1	GND
0.721500	27.60	10.2	46	18.4	AV	L1	GND
1.684500	29.70	10.2	46	16.3	AV	L1	GND
2.116500	27.40	10.2	46	18.6	AV	L1	GND
12.201000	23.10	10.5	50	26.9	AV	L1	GND
19.441500	27.70	10.5	50	22.3	AV	L1	GND

Polarization	N



MEASUREMENT RESULT: "GM2107295061_fin"

7/29/2021 6:23PM

Frequency MHz	Level dBµV	Transd dB	Limit dBµV	Margin dB	Detector	Line	PE
0.163500	53.50	10.2	65	11.8	QP	N	GND
0.505500	34.40	10.2	56	21.6	QP	N	GND
1.239000	34.20	10.2	56	21.8	QP	N	GND
2.319000	30.90	10.2	56	25.1	QP	N	GND
7.831500	23.60	10.3	60	36.4	QP	N	GND
26.281500	33.50	10.5	60	26.5	QP	N	GND

MEASUREMENT RESULT: "GM2107295061_fin2"

7/29/2021 6:23PM

Frequency MHz	Level dBµV	Transd dB	Limit dBµV	Margin dB	Detector	Line	PE
0.199500	33.50	10.2	54	20.1	AV	N	GND
0.555000	27.20	10.2	46	18.8	AV	N	GND
1.189500	27.00	10.2	46	19.0	AV	N	GND
2.305500	25.40	10.2	46	20.6	AV	N	GND
5.923500	17.90	10.2	50	32.1	AV	N	GND
19.401000	29.60	10.5	50	20.4	AV	N	GND

5.1.3. Harmonic Current Emission

LIMIT

EN61000-3-2 Clause 7

- **Class A equipment**

Harmonic order n	Maximum permissible harmonic current A
Odd harmonics	
3	2,30
5	1,14
7	0,77
9	0,40
11	0,33
13	0,21
15 ≤ n ≤ 39	$0{,}15 \frac{15}{n}$
Even harmonics	
2	1,08
4	0,43
6	0,30
8 ≤ n ≤ 40	$0{,}23 \frac{8}{n}$

- **Class B equipment**

not exceed the values givenin Class A limit multiplied by a factor of 1,5

- **Class C equipment**

Active input power >25 W

Harmonic order n	Maximum permissible harmonic currrent expressed as a percentage of the input current at the fundamental frequency %
2	2
3	$30 \cdot \lambda$ *
5	10
7	7
9	5
11 ≤ n ≤ 39 (odd harmonics only)	3
* λ is the circuit power factor	

Active input power ≤25 W

Harmonic order n	Maximum permissible harmonic current per watt mA/W	Maximum permissible harmonic current A
3	3,4	2,30
5	1,9	1,14
7	1,0	0,77
9	0,5	0,40
11	0,35	0,33
13 ≤ n ≤ 39 (odd harmonics only)	$\frac{3{,}85}{n}$	See Table 1

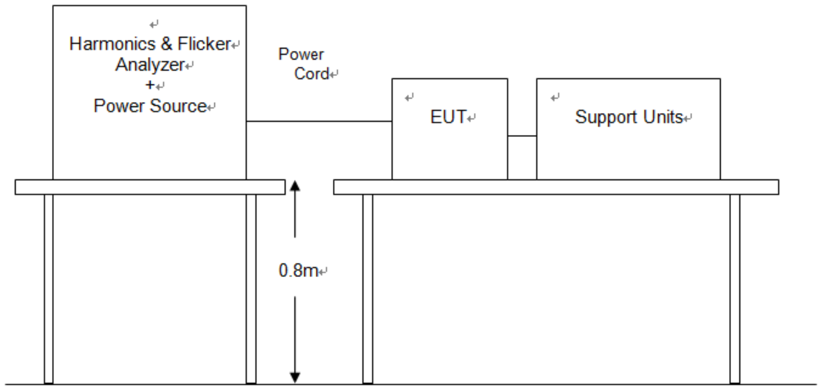
or

the third harmonic current, expressed as a percentage of the fundamental current,shall not exceed 86 % and the fifth harmonic current shall not exceed 61 %. Also, thewaveform of the input current shall be such that it reaches the 5 % current thresholdbefore or at 60º, has its peak value before or at 65º and does not fall below the 5 %current threshold before 90º, referenced to any zero crossing of the fundamentalsupply voltage. The current threshold is 5 % of the highest absolute peak value thatoccurs in the measurement window, and the phase angle measurements are made onthe cycle that includes this absolute peak value

➢ **Class D equipment**

Harmonic order n	Maximum permissible harmonic current per watt mA/W	Maximum permissible harmonic current A
3	3,4	2,30
5	1,9	1,14
7	1,0	0,77
9	0,5	0,40
11	0,35	0,33
$13 \le n \le 39$ (odd harmonics only)	$\frac{3,85}{n}$	See Table 1

TEST CONFIGURATION



TEST PROCEDURE

Please refer to EN61000-3-2 for the measurement methods.

TEST RESULTS

☐ **Passed** ☒ **Not Applicable**

Note: The power of the EUT is less than 75W, So this test item is not applicable.

5.1.4. Voltage Fluctuation and Flicker

LIMIT

Please refer to EN61000-3-3

– the value of Pst shall not be greater than 1,0;

– the value of Plt shall not be greater than 0,65;

– the value of d(t) during a voltage change shall not exceed 3,3 % for more than 500 ms;

– the relative steady-state voltage change, dc, shall not exceed 3,3 %;

– the maximum relative voltage change dmax, shall not exceed

a) 4 % without additional conditions;

b) 6 % for equipment which is:

– switched manually, or

– switched automatically more frequently than twice per day, and also has either adelayed restart (the delay being not less than a few tens of seconds), or manualrestart, after a power supply interruption.

c) 7 % for equipment which is

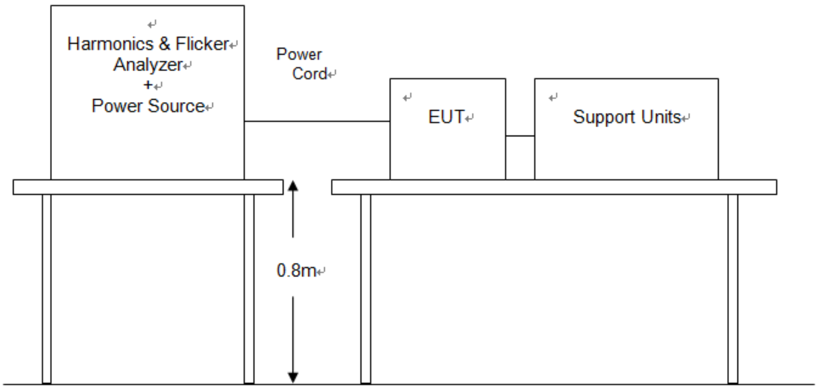
– attended whilst in use (for example: hair dryers, vacuum cleaners, kitchenequipment such as mixers, garden equipment such as lawn mowers, portable toolssuch as electric drills), or

– switched on automatically, or is intended to be switched on manually, no more thantwice per day, and also has either a delayed restart (the delay being not less than afew tens of seconds) or manual restart, after a power supply interruption.

In the case of equipment having several separately controlled circuits in accordance with 6.6,limits b) and c) shall apply only if there is delayed or manual restart after a power supplyinterruption; for all equipment with automatic switching which is energised immediately onrestoration of supply after a power supply interruption, limits a) shall apply; for all equipmentwith manual switching, limits b) or c) shall apply depending on the rate of switching.

Pst and Plt requirements shall not be applied to voltage changes caused by manual switching.

TEST CONFIGURATION



TEST PROCEDURE

Please refer to EN61000-3-3 for the measurement methods.

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Standard used:	EN/IEC 61000-3-3 Flicker
Short time (Pst):	10min
Observation time:	120min (12 Flicker measurements)
Flickermeter:	230V / 50Hz according IEC 61000-4-15 Ed.2
Flicker Impedance:	Zref (IEC 60725)

Test Result	PASS

Maximum Flicker results

	EUT values	**Limit**	**Result**
Pst	0.028	1.00	PASS
Plt	0.028	0.65	PASS
dc [%]	0.018	3.30	PASS
dmax [%]	0.082	4.00	PASS
dt [s]	0.000	0.50	PASS

Detail Flicker data

Flicker measurement 1	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.018	3.30	PASS
dmax [%]	0.082	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 2	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.013	3.30	PASS
dmax [%]	0.054	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 3	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.017	3.30	PASS
dmax [%]	0.047	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 4	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.017	3.30	PASS
dmax [%]	0.046	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 5	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.011	3.30	PASS
dmax [%]	0.045	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 6	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.007	3.30	PASS
dmax [%]	0.047	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 7	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.049	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 8	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.042	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 9	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.013	3.30	PASS
dmax [%]	0.049	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 10	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.040	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 11	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.048	4.00	PASS
dt [s]	0.000	0.50	PASS

Flicker measurement 12	**EUT values**	**Limit**	**Result**
Pst	0.028	1.00	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.045	4.00	PASS
dt [s]	0.000	0.50	PASS

5.2. IMMUNITY

Performance criteria

- ***EN301489-3:***
  - *Performance criterion A applies for immunity tests with phenomena of a continuous nature;*
  - *Performance criterion B applies for immunity tests with phenomena of a transient nature.*

NOTE: Whether a phenomenon is considered transient, continuous or otherwise is indicated in the test procedures for the phenomenon in ETSI EN 301 489-1 [1], clause 9.

Criteria	*During test*	*After test*
A	*Operate as intended* *No loss of function* *No unintentional responses*	*Operate as intended* *No loss of function* *No degradation of performance* *No loss of stored data or user programmable functions*
B	*May show loss of function* *No unintentional responses*	*Operate as intended* *Lost function(s) shall be self-recoverable* *No degradation of performance* *No loss of stored data or user programmable functions*

5.2.1. Electrostatic Discharge

PERFORMANCE CRITERION

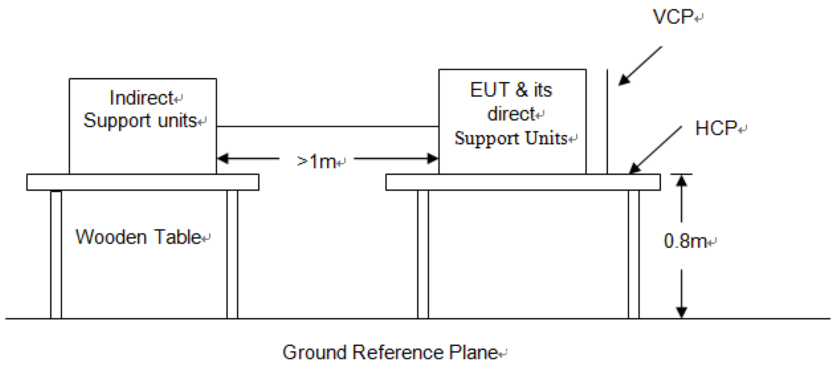
Standard	Criterion
ETSI EN301489-3	Criteria B

TEST LEVEL

Contact Discharge at ±2kV,±4kV;

Air Discharge at±2kV,±4kV,±8kV

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.3.2 and EN 61000-4-2 for the measurement methods.

Contact Discharge:

The ESD generator is held perpendicular to the surface to which the discharge is applied and the tip of the discharge electrode touch the surface of EUT. Then turn the discharge switch. The generator is then re-triggered for a new single discharge and repeated at least 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Air Discharge:

Air discharge is used where contact discharge can't be applied.The round discharge tip of the discharge electrode shall be approached as fast as possible to touch the EUT. After each discharge, the discharge electrode shall be removed from the EUT. The generator is then re-triggered for a new single discharge and repeated at least 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Indirect discharge for horizontal coupling plane:

At least 10 single discharges shall be applied to the horizontal coupling plane, at points on each side of the EUT.

Indirect discharge for vertical coupling plane:

At least 10 single discharges shall be applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, is placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges shall be applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

TEST MODE

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicabl**

Test mode		O*			
Type	Type of discharge	Discharge voltage (kV)	Observations Performance	CriteriaLevel	Result
Direct	Contact discharge	±2	N/A	N/A	Pass
		±4	N/A	N/A	
	Air discharge	±2	N/A	N/A	
		±4	N/A	N/A	
		±8	N/A	N/A	
Indirect	HCP (6 sides)	±2	No degradation in performance of the EUT was observed (A)	B	Pass
		±4	A	B	
	VCP (4 sides)	±2	A	B	
		±4	A	B	

Note:

The ancillary equipment's specification for an acceptable level of performance or degradation of performance during and/or after the ESD tests.

Contact discharge-Yellow, Air discharge-Red

5.2.2. Radio Frequency Electromagnetic Field

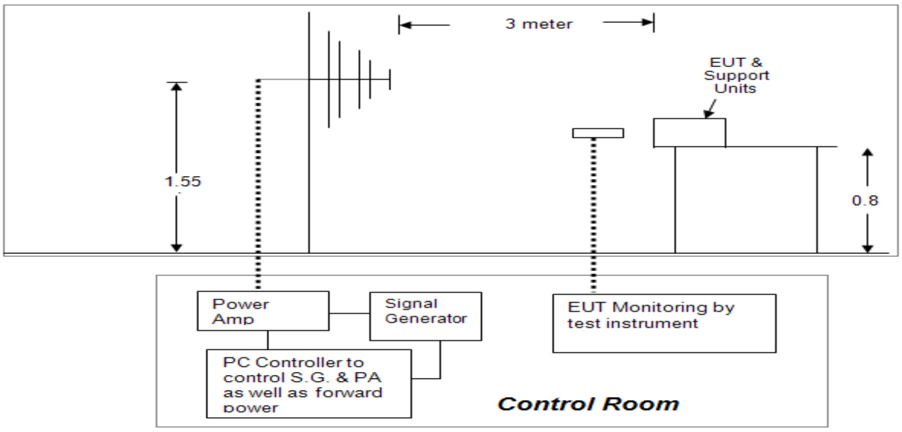
PERFORMANCE CRITERION

Standard	Criterion
ETSI EN301489-3	Criteria A

TEST LEVEL

Test frequency range: 80MHz~6000MHz
Level: 3V/m (80%, 1kHz Amplitude Modulation)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.2.2 and EN 61000-4-3 for the measurement methods.

TEST MODE

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Test mode:	O*		
Antenna Polarity	Observations (Performance Criterion)	CriteriaLevel	Result
H/V	*No degradation in performance of the EUT was observed* (A)	A	Pass

5.2.3. Fast Transients Common Mode

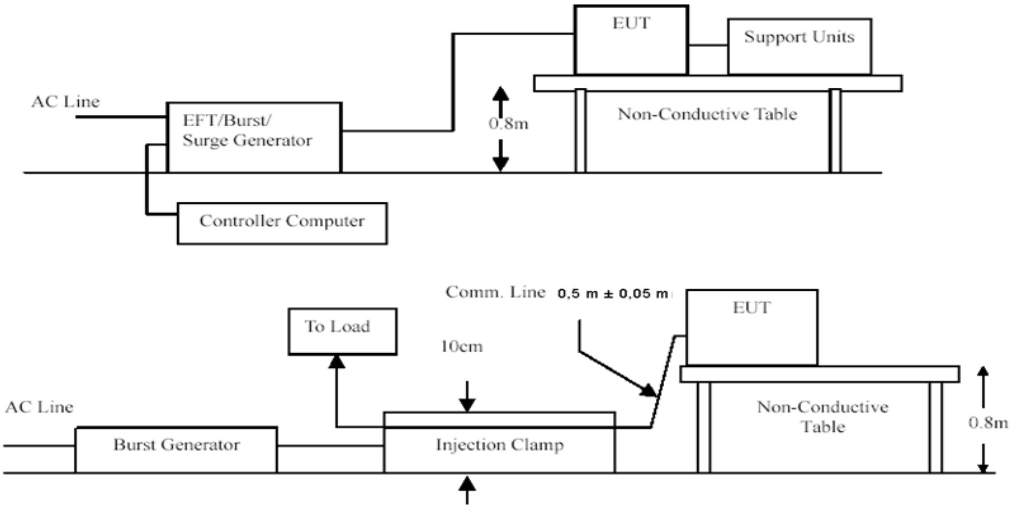
PERFORMANCE CRITERION

Standard	Criterion
ETSI EN301489-3	Criteria B

TEST LEVEL

Level: 1KV for AC port, 0.5KV for signal port
Impulse Frequency: 5 kHz;
Tr/Td: 5/50ns;
Burst Duration: 15ms; Burst Period: 3Hz

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.4.2and EN 61000-4-4 for the measurement methods.

TEST MODE

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Test mode:		O*		
Lead under Test	Coupling Direct / Clamp	Observations (Performance Criterion)	CriteriaLevel	Result
L	Direct	*No degradation in performance of the EUT was observed (A)*	B	Pass
N	Direct	A	B	Pass
L-N	Direct	A	B	Pass

5.2.4. Surge

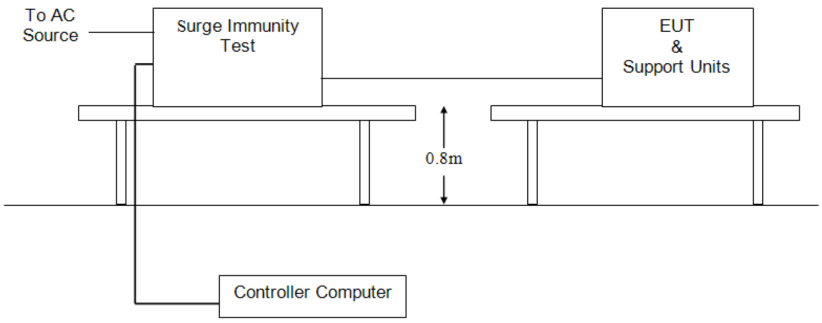
PERFORMANCE CRITERION

Standard	Criterion
ETSI EN301489-3	Criteria B

TEST LEVEL

Level: 1kV for line to line, 2kV for line to ground
Voltage Waveform: 1.2/50 us; Current Waveform: 8/20 us
Pluse quantity: 5, interval time: 60 seconds
Phase: 0°, 90°, 180°, 270°

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.8.2and EN 61000-4-5 for the measurement methods.

TEST MODE

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Test mode:	O*			
Lead under Test	Phase	Observations (Performance Criterion)	CriteriaLevel	Result
L-N	0°	*No degradation in performance of the EUT was observed* (A)	B	Pass
	90°	A	B	Pass
	180°	A	B	Pass
	270°	A	B	Pass

5.2.5. Radio Frequency Common Mode

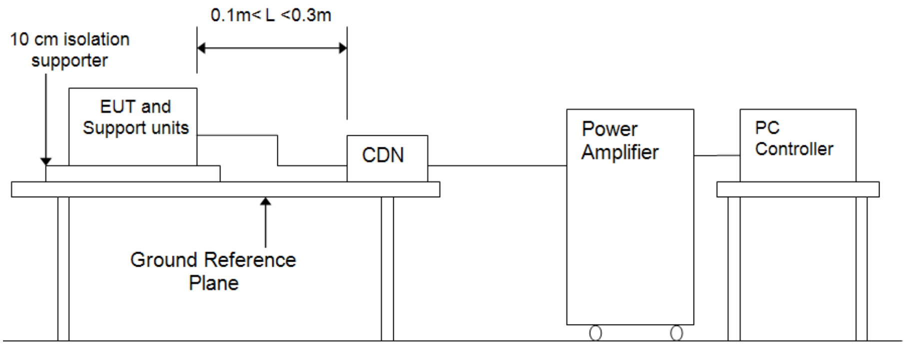
PERFORMANCE CRITERION

Standard	Criterion
ETSI EN301489-3	Criteria A

TEST LEVEL

Test frequency range: 150kHz~80MHz
Level: 3Vrms (80%, 1kHz Amplitude Modulation)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.5.2and EN 61000-4-6 for the measurement methods.

TEST MODE

Please refer to the Clause 3.4

TEST RESULTS

☒ **Passed** ☐ **Not Applicable**

Test mode:	O*		
Injected Position	Observations (Performance Criterion)	CriteriaLevel	Result
AC Mains	*No degradation in performance of the EUT was observed. (A)*	CR	Pass

5.2.6. Voltage Dips and Interruptions

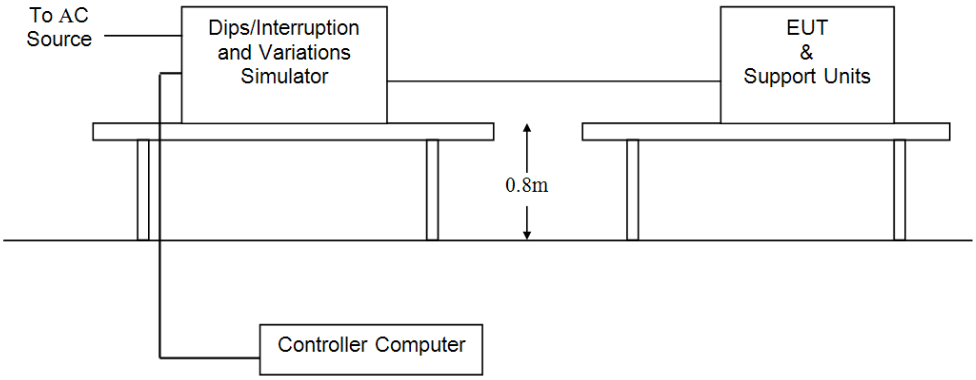
PERFORMANCE CRITERION

Standard	Criterion
ETSI EN301489-3	Criteria B for voltage dip Criteria C for voltage interruption

TEST LEVEL

0% of VT(Supply Voltage) for 0.5 period
0% of VT(Supply Voltage) for 1.0 period
70% of VT(Supply Voltage) for 25 period
0% of VT(Supply Voltage) for 250 period
Dip quantity: 3, interval time: 10 seconds

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.7.2and EN 61000-4-11 for the measurement methods.

TEST MODE

Please refer to the Clause 3.4

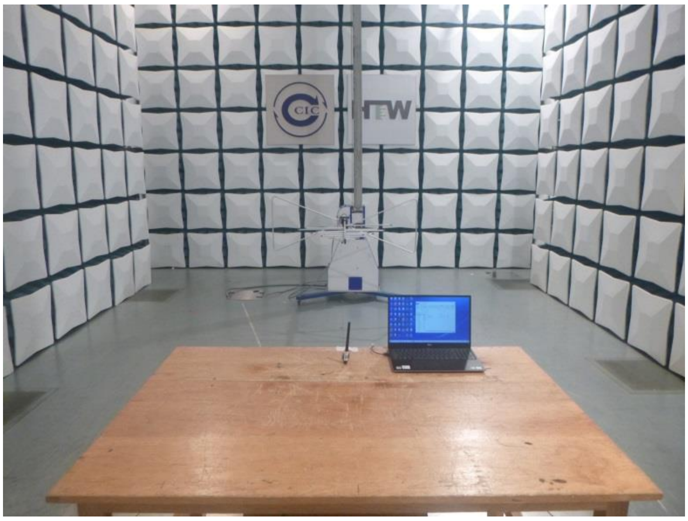
TEST RESULTS

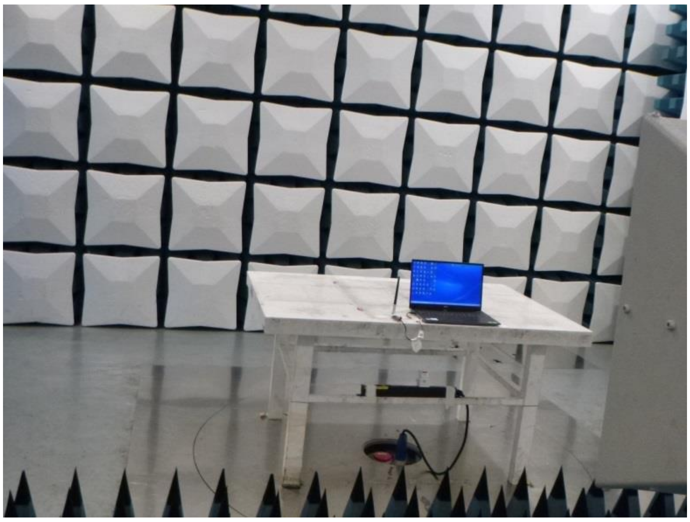
☒ **Passed** ☐ **Not Applicable**

Test mode:		O*			
Test Voltage %	Duration periods	Phase angle	Observations (Performance Criterion)	CriteriaLevel	Result
0	0.5	0°, 90°, 180°, 270°	*No degradation in performance of the EUT was observed. (A)*	B	Pass
0	1.0	0°, 90°, 180°, 270°	A	B	Pass
70	25	0°, 90°, 180°, 270°	A	B	Pass
0	250	0°, 90°, 180°, 270°	*During the test, the power shut down, after the experiment, the function can automatically return to normal. (B)*	C	Pass

6. Test Setup Photos of the EUT

Radiated Emission



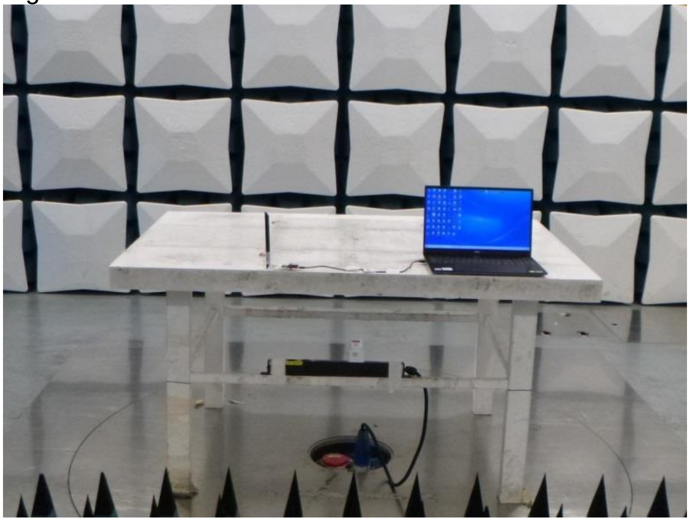


Conducted Emission (AC Mains)

Harmonic Current Emission / Voltage Fluctuation and Flicker

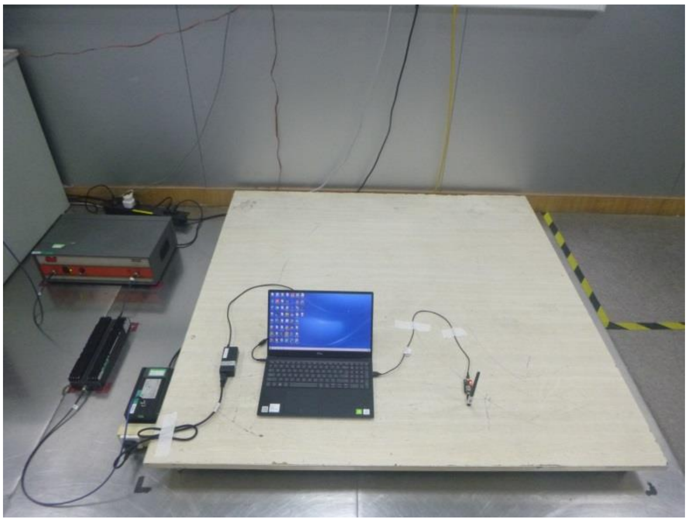
Electrostatic Discharge

Radio Frequency Electromagnetic Field



Fast Transients- Common Mode / Surge / Voltage Dips and Interruptions

Radio Frequency Common Mode



7. External and Internal Photos of the EUT

Reference to the test report No. CHTEW21100065

-----------------*End of Report*----------------